

IV SEMESTER

STATISTICS AND PROBABILITY FOR DATA SCIENCE	
Course Code: AD41	Credits: 2:1:0
Pre – requisites: Basic Probability	Contact Hours: 28L+14T
Course Coordinator: Dr. S H C V SUBBA BHATTA	

Course Content

Unit I

Statistics: Curve fitting by the method of least squares, fitting linear, quadratic and geometric curves, correlation, Rank correlation.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: https://onlinecourses.nptel.ac.in/noc23_ma50/preview

Unit II

Regression Analysis: Linear regression, Multiple correlation and regression, Partial correlation and regression, ridge regression, logistic regression, Review of probability, Probability interpretation of correlation and regression.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: https://onlinecourses.nptel.ac.in/noc23_ma50/preview

Unit III

Probability Distributions I: Random variables, Binomial distribution, Poisson distribution, Uniform distribution, Exponential distribution, Gamma distribution and Normal distribution.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/117105085>
<http://nitttrc.edu.in/nptel/courses/video/111105041/L19.html>

Unit IV

Probability Distributions II: Lognormal distribution, Central limit theorem, Relation between Binomial and Normal distribution, Relation between Poisson and Normal distribution.

Joint probability distribution: Joint probability distribution (both discrete and

continuous), conditional probability, conditional expectation.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://www.nptelvideos.com/course.php?id=746>

Unit V

Sampling and Statistical Inference: Sampling distributions, concepts of standard error and confidence interval, level of significance, type I and type II errors, one tailed and two tailed tests, Z-test: for single mean, for single proportion and for difference between means, Student's t –test: for single mean and for difference between two means, F – test for equality of two variances, Chi-square test: for goodness of fit and for independence of attributes.

- Pedagogy/Course delivery tools: Chalk and talk, PowerPoint Presentation, Videos
- Links: <http://www.nitttrc.edu.in/nptel/courses/video/111105041/L23.html>

Suggested Learning Resources

Text Books:

1. R.E. Walpole, R. H. Myers, R. S. L. Myers and K. Ye – Probability and Statistics for Engineers and Scientists – Pearson Education – Delhi – 9th edition – 2012.
2. T. Veerarajan- Probability, Statistics and Random processes – Tata McGraw-Hill Education – 3rd edition -2017.
3. Johnson/Miller: Miller & Freund's Probability and Statistics for Engineers, Ninth Edition, Pearson Education India -2018.

Reference Books:

1. Sheldon M. Ross – Probability models for Computer Science – Academic Press, 10th edition Elsevier– 2010.
2. Murray R Spiegel, John Schiller & R. Alu Srinivasan – Probability and Statistics – Schaum's outlines -4th edition-2012.
3. Kishore S. Trivedi – Probability & Statistics with Reliability, Queuing and Computer Science Applications – John Wiley & Sons – 2nd edition – 2008.
4. Dirk P. Kroese, Zdravko I. Botev, Thomas Taimre, Radislav Vaisman, Data Science and Machine Learning, Mathematical and Statistical Methods, Chapman and Hall/CRC, 2019.

Course Outcomes (COs):

At the end of the course, students will be able to

1. Apply method of least squares to find the curve of best fit for the given data. (PO-1, 2 & PSO-1,2)
2. Understand the differences between simple and multiple linear regression. (PO-1, 2 & PSO-1,2)
3. Analyze the given random data and its probability distributions. (PO-1, 2 & PSO-1,2).
4. Find parameters of Continuous Probability distributions and calculate the marginal and conditional distributions of bivariate random variables. (PO-1, 2 & PSO-1,2).
5. Choose an appropriate test of significance and make inference about the population from a sample. (PO-1, 2 & PSO-1,2).

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tool	Marks	Course outcomes addressed
Internal test-I	30	CO1, CO2, CO3
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other components	Marks	Course outcomes addressed
Quiz	10	CO1, CO2, CO3
Assignment	10	CO3, CO4, CO5
Semester End Examination (SEE):	100	CO1, CO2, CO3, CO4, CO5